



CONTROL UPTO 16 APPLIANCES

I am very impressed by the 'Control Upto 16 Appliances Without Using a Micro-controller' DIY article published in February 2019 issue. However, MM74C922 key encoder used in the project is not easily available. Also, 4-to-16 decoder (74HC154) seems obsolete. Are there any alternative or compatible ICs available for the project?

A. Babu Narendra Prasad

The author Sanjay Kumar Gupta replies: You may buy these ICs from the following websites:

<https://hacktronics.co.in/74-series-ic/mm74c922n-fsc-encoder-74c922-16-key-matrix-keyboard-encoder-ic> and <https://www.aliexpress.com/w/wholesale-mm74c922>

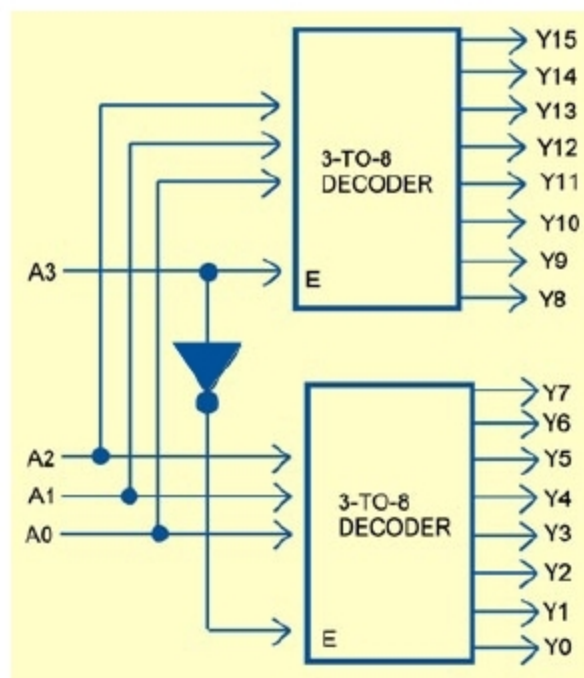


Fig. 1: Equivalent 4-to-16 decoder circuit

If you are unable to get this decoder IC, you can use two 3-to-8 decoder ICs (74LS138) to get equivalent output of 4-to-16 decoder, as shown in Fig. 1.

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WIRELESS WATER-LEVEL INDICATOR-CUM-CONTROLLER

I am interested in the 'Stop Tank Overflow With This Wireless Water-Level Indicator-Cum-Controller' DIY article

From <https://electronicsforu.com>

Electronics Projects

The 'Home Automation System Using a Simple Android App' DIY project available on your website, <https://electronicsforu.com>, is a good project.

Prasanna K.

This is regarding the 'Power Bank for Smartphones' article available on your website. I have built the power bank in accordance with the given circuit. The circuit is working but charging is very slow. I am using four 18650 4300mAh batteries in parallel. What can I do to improve performance of the circuit?

Yasar

The author T.K. Hareendran replies: It seems you are using four 18650 4300mAh lithium-ion cells in parallel. That is certainly not possible in this circuit, especially with TP4056. Perhaps, you will get a good insight through the following datasheet: <https://dlnmh9ip6v2uc.cloudfront.net/datasheets/Prototyping/TP4056.pdf>

In the 'RFID-Based Access Control Using Arduino' DIY project available on your website, will it be possible to add

another lock using another RFID tag?
David

EFY. Yes, definitely! With the same RFID reader and Arduino, you can add another RFID tag in the source code of Arduino, along with a solenoid driver.

In the 'Smoke-, Alcohol- and LPG-Detection Alarm' DIY article available on your website, can I use MQ5 without making changes in the circuit?

Vishnu

EFY. Yes, you can use MQ5 without any modifications in the circuit! Note that, MQ5 is not meant for alcohol detection but is mainly used for LPG and natural gas detection.

In the 'Designing Wireless Gas Detection and Monitoring Systems' DIY article available on your website, the `strcat()` function in the code is not executing. Please help me include string function in the code to display text messages.

Mahalaxmi

The author Dogin Ibrahim replies: It is possible that C-String in the source is not enabled in Library Manager.

published in September 2019 issue. How is the signal transmitted from the transmitter unit to the receiver unit? Is this project based on programming?

Kiran

The author Arup Kumar Sen replies: You may refer to the data sheet of HT12E. It is commonly-used for radio frequency (RF) applications. By using the paired HT12E encoder and HT12D decoder, we can easily transmit and receive twelve bits of parallel data serially. HT12E simply converts twelve-bit parallel data to serial output, which can be transmitted through an RF transmitter. So, there is no programming required.

FM TRANSMITTER

I have a doubt in the 'Make Your Own FM Transmitter For Broadcasting' DIY article published in November 2019 issue. Can I use 8V power supply for varactor 1SV149 as per its datasheet? Also, can I replace 1SV149 with another equivalent varactor?

A. Bhatt

The author Joy Mukherji replies: It works fine in my circuit with 12V supply even though maximum rating is only 8V. You can replace it with any other varactor diode, which is easily available. You can also try using MV2109 in place of 1SV149.